

EXPERIMENT 1:

AIM:

To display the unique department IDs from the employee dataset using Pandas.

DATASET:

employee_id	first_name	department_id
101	John	10
102	Smith	20
103	David	10
104	Sara	30
105	Tom	20

PROGRAM:

```
exp1.py > ...
1 import pandas as pd
2 df = pd.read_excel(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp1_data.xlsx")
3 print("Distinct Department IDs:")
4 print(df["department_id"].drop_duplicates())
```

OUTPUT:

```
\Python313\python.exe "c:/Users/dhars/Documents/Query processing/lab_questions/exp1.py"
Distinct Department IDs:
0    10
1    20
3    30
Name: department_id, dtype: int64
PS C:\Users\dhars\Documents\Query processing\lab_questions> 
```

RESULT:

The distinct department IDs were successfully extracted from the employees dataset using Pandas. The unique department IDs displayed were **10, 20, and 30**.

EXPERIMENT 2:

AIM:

To display employee IDs that appear two or more times in the job history dataset.

INPUT:

employee_id	job_id
101	IT_PROG
101	SA_REP
102	HR_REP
103	MK_MAN
103	FI_ACC
103	AD_ASST

PROGRAM:

```
exp2.py > ...
1 import pandas as pd
2 df = pd.read_excel(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp2_data.xlsx")
3 result = df["employee_id"].value_counts()
4 print("Employees who have done two or more jobs:")
5 print(result[result >= 2])
```

OUTPUT:

```
PS C:\Users\dhars\Documents\Query processing\lab_questions> & C:\Users\dhars\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/dhars/Documents/Query processing/lab_questions/exp2.py"
Employees who have done two or more jobs:
employee_id
103      3
101      2
Name: count, dtype: int64
```

RESULT:

The employee IDs who have worked in **two or more jobs** were successfully identified using Pandas. Employees **101** and **103** were found to have worked in multiple jobs.